

| INVESTIGATION CARD                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | DISK-01 |
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| <b>DISK IMAGE &amp; ANALYSIS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |
| TECHNIQUE: Forensic Disk Imaging & Analysis                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |
| MITRE ATTACK: T1005 (Data from Local System), T1025 (Data from Removable Media)                                                                                                                                                                                                                                                                                                                                                                                                                  |         |
| DIFFICULTY CLASS: 12                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |
| BUDGET COST: 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |         |
| DURATION: 2 turns (v2.2 rush: pay +5 Budget for Duration 1)                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |
| DESCRIPTION<br>Create a bit-for-bit disk image of the compromised system, then examine file system artifacts, deleted files, and hidden data. This is a foundational forensic technique. What You're Looking For: - Malware files (executables, scripts, libraries) - Deleted files (file carving reveals overwritten data) - Persistence mechanisms (startup folders, registry runs) - Downloaded files (browser cache, temp directories) - Suspicious file timestamps (backdating, mismatches) |         |
| SUCCESS (ROLL ≥ DC 12)<br>Discover ONE Evidence card from: Malware Sample, Persistence Mechanism, or Downloaded Malware evidence set. Advance (only if no Evidence card produced):<br>Timeline Completeness +10%, Attack Chain +15% PARTIAL SUCCESS (roll DC-2 to DC-1 = 10-11); Discover INCOMPLETE Evidence card (partial findings). Advance: Timeline Completeness +5%, Attack Chain +5% FAILURE (roll < 10): No evidence discovered. Budget still spent. Take a turn.                        |         |
| CHAIN OF CUSTODY<br>Disk image must be bit-for-bit copy. Chain of custody: Strong - Imaging is gold standard in forensics. / All evidence from this source is admissible in court<br>SKILL MODIFIERS: +2 if investigator has formal GC/HQ/CFE training +1 if investigator has IT administration background +1 if team provides detailed explanation of imaging process                                                                                                                           |         |
| NOTES<br>• Disk imaging is time-consuming (hence 2 turn cost) • Can be combined with DISK-02 for deeper analysis • Foundation for all disk-based forensic work • Works best on traditional disk systems (less effective on SSDs with wear-leveling and TRIM commands)                                                                                                                                                                                                                            |         |
| INCIDENT ZERO - FORENSICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |         |

| INVESTIGATION CARD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | MEM-02 |
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| <b>MEMORY FORENSICS DEEP DIVE</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |
| TECHNIQUE: Advanced Volatile Memory Analysis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |
| MITRE ATTACK: T1112 (Modify Registry), T1055 (Process Injection), T1140 (Deobfuscate/Decode Files)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |
| DIFFICULTY CLASS: 15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |
| BUDGET COST: 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |
| DURATION: 3 turns (expert-level analysis)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |
| DESCRIPTION<br>Expert-level memory analysis including malware behavior simulation, deobfuscation of shellcode, reverse engineering of code injected into memory, and recovery of encryption keys. Requires deep expertise in assembly language, malware tactics, and memory layouts. What You're Looking For: - Obfuscated/encrypted malware payloads (deobfuscate them) - Code injection techniques (understand HOW malware hides) - Encryption keys and passphrases in memory (crypto recovery) - Malware command history (recent attacker commands) - Process hollowing or code caves (anti-analysis techniques) - Privilege escalation vulnerabilities in use |        |
| SUCCESS (ROLL ≥ DC 15)<br>Discover TWO Evidence cards: One from malware behavior set (e.g., Encryption Keys, Command History) + one from attack technique set (e.g., Code Injection Method, Exploitation Used). Advance (only if no Evidence card produced): Attack Chain +25%, Attribution +20% PARTIAL SUCCESS (roll 13-14); Discover ONE complete Evidence + incomplete second evidence. Advance: Attack Chain +15%, Attribution +10% FAILURE (roll < 13): Malware uses sophisticated anti-analysis; analysis fails.                                                                                                                                           |        |
| CHAIN OF CUSTODY<br>Analysis documentation critical (how did you reach conclusions) Chain of Custody: Strong if reverse engineering is documented. / Conclusions must be explained clearly for court admissibility                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |
| SKILL MODIFIERS<br>+3 if investigator has GCFA (Certified Forensic Analyst) +2 if malware reverse engineering background +1 if Malware Analysis card already completed +1 if detailed explanation of deobfuscation approach +2 if malware is heavily obfuscated or virtualized                                                                                                                                                                                                                                                                                                                                                                                    |        |
| CONTINUED IN MODULE RULES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |
| INCIDENT ZERO - FORENSICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |

| INVESTIGATION CARD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | NET-01 |
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| <b>NETWORK TRAFFIC ANALYSIS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |
| TECHNIQUE: Packet Capture & Flow Analysis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |
| MITRE ATTACK: T1041 (Exfiltration Over C2), T1048 (Alternative Protocol), T1071 (Application Layer Protocol)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |
| DIFFICULTY CLASS: 12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |
| BUDGET COST: 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |
| DURATION: 2 turns                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |
| DESCRIPTION<br>Analyze network traffic captures (PCAP) or network flow records (NetFlow) to identify communication patterns, exfiltration evidence, and command-and-control connections. Uses tools like Wireshark, Zeek, or commercial traffic analysis platforms. What You're Looking For: - Unusual outbound connections (C2 domains, IPs) - Large data transfers (exfiltration evidence) - Encrypted tunnels (VPN, proxy connections) - DNS queries for suspicious domains - HTTP user agents inconsistent with legitimate software - Beacon-like patterns (regular connection attempts) |        |
| SUCCESS (ROLL ≥ DC 12)<br>Discover ONE Evidence card from: C2 Server Evidence, Exfiltration Traffic Pattern, or Suspicious Domain Lookup. Advance (only if no Evidence card produced): Attack Chain +20%, Attribution +15% PARTIAL SUCCESS (roll 10-11); Discover suspicious traffic but destination unclear. Advance: Attack Chain +10%, Attribution +5% FAILURE (roll < 10): Traffic too encrypted or obfuscated; cannot analyze.                                                                                                                                                          |        |
| CHAIN OF CUSTODY<br>PCAP files must include timestamp and collection metadata. Chain of Custody: Strong if collected from router/IDS. / Admissible (widely accepted for network evidence) - Encrypted traffic reveals patterns but not content                                                                                                                                                                                                                                                                                                                                               |        |
| SKILL MODIFIERS<br>+2 if investigator has Wireshark/packet analysis certification +1 if network engineering background +1 if Threat Attribution evidence already discovered -1 if traffic is heavily encrypted or anonymized                                                                                                                                                                                                                                                                                                                                                                 |        |
| NOTES<br>• Reveals attacker communication patterns • Can identify C2 infrastructure • Exfiltration volume is critical evidence • Encrypted traffic is harder to analyze but patterns visible                                                                                                                                                                                                                                                                                                                                                                                                 |        |
| INCIDENT ZERO - FORENSICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |

| INVESTIGATION CARD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | DISK-02 |
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| <b>FILE SYSTEM CARVING</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |         |
| TECHNIQUE: Advanced File Recovery & Data Carving                                                                                                                                                                                                                                                                                                                                                                                                                                                   |         |
| MITRE ATTACK: T1074 (Data Staged), T1465 (Data Destruction)                                                                                                                                                                                                                                                                                                                                                                                                                                        |         |
| DIFFICULTY CLASS: 14                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |         |
| BUDGET COST: 15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |
| DURATION: 3 turns (specialized expertise required)                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |
| DESCRIPTION<br>Advanced carving techniques recover data from unallocated disk space and file slack, even when files have been deleted and storage sectors overwritten. Uses specialized tools like Encase, FTK, or open-source carving tools. What You're Looking For: - Deleted malware (recovered from free space) - Temporary files (attacker staging data before exfil) - Encryption keys or passphrases (memory remnants on disk) - Hidden partitions or file systems - Slack space artifacts |         |
| SUCCESS (ROLL ≥ DC 14)<br>Discover ONE Evidence card from: Deep Malware Samples, Encryption Keys Found, or Hidden Backdoor Evidence. Advance (only if no Evidence card produced): Attack Chain +20%, Chain of Custody +10% PARTIAL SUCCESS (roll 12-13); Discover partial data (e.g., fragments of deleted file). Advance: Attack Chain +10%, Chain of Custody +5% FAILURE (roll < 12): Data too corrupted or already overwritten. No recovery.                                                    |         |
| CHAIN OF CUSTODY<br>Carving is technically sound but must be documented carefully. Chain of Custody: Strong if done by certified analyst. / Partial carving may be challenged in court (incomplete file recovery). Recommend combining with other techniques.                                                                                                                                                                                                                                      |         |
| SKILL MODIFIERS<br>+2 if investigator has GCFE (Certified Forensic Examiner) +1 if investigator has disk forensics experience +1 if combined with DISK-01 investigation already done -1 if SSD drives present (wear-leveling complicates carving)                                                                                                                                                                                                                                                  |         |
| NOTES<br>• Expensive investigation (15 budget) for specialized work • Can take weeks in real incidents; represented as 3 turns • Most valuable for discovering deleted persistence and encryption keys • Less effective on modern systems with TRIM/wear-leveling                                                                                                                                                                                                                                  |         |
| INCIDENT ZERO - FORENSICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |         |

| INVESTIGATION CARD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | LOG-01 |
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| <b>EVENT LOG ANALYSIS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |
| TECHNIQUE: Windows/Linux Log Examination                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |
| MITRE ATTACK: T1552 (Unsecured Credentials), T1098 (Account Manipulation) T1021 (Remote Services), T1078 (Valid Accounts)                                                                                                                                                                                                                                                                                                                                                                                         |        |
| DIFFICULTY CLASS: 11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |
| BUDGET COST: 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |
| DURATION: 1 turn                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |
| DESCRIPTION<br>Analyze system event logs (Windows Event Log, syslog, etc.) to identify user logins, privilege escalations, file access, and process execution. This is foundational and relatively quick-useful for establishing a basic timeline. What You're Looking For: - Failed login attempts (brute force evidence) - Successful logins from unusual locations/times - Privilege escalation attempts (Runs, sudo) - Process creation events / Service installation events - File access to sensitive files |        |
| SUCCESS (ROLL ≥ DC 11)<br>Discover ONE Evidence card from: Suspicious Login Timeline, Privilege Escalation Attempt, or Service Installation evidence. Advance (only if no Evidence card produced): Timeline Completeness +15%, Attack Chain +10% PARTIAL SUCCESS (roll 9-10); Discover partial timeline (logs are fragmented or unclear). Advance: Timeline Completeness +5%, Attack Chain +5% FAILURE (roll < 9): Logs were deleted or corrupted; no useful evidence.                                            |        |
| CHAIN OF CUSTODY<br>Logs must be exported with metadata (timestamps, user context). Chain of Custody: Strong if logs are digitally signed. / Admissible in court (widely accepted evidence type)                                                                                                                                                                                                                                                                                                                  |        |
| SKILL MODIFIERS<br>+1 if investigator has Windows administration experience +1 if investigator has SIEM/log analysis background +1 if detailed explanation of log analysis approach +2 if prior Timeline Reconstruction investigation completed                                                                                                                                                                                                                                                                   |        |
| NOTES<br>• Cheapest investigation (5 budget) - good starting point • Fastest (1 turn) - can be done early in investigation • Foundation for Timeline Reconstruction and Log Correlation • May be ineffective if attacker deleted logs (add anti-forensics penalty: +2 DC)                                                                                                                                                                                                                                         |        |
| INCIDENT ZERO - FORENSICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |

| INVESTIGATION CARD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | NET-02 |
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| <b>PACKET CAPTURE DEEP ANALYSIS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |
| TECHNIQUE: Advanced Protocol Forensics & Reconstruction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |
| MITRE ATTACK: T1557 (Adversary-in-the-Middle), T1040 (Network Sniffing), T1071 (Application Layer Protocol)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |
| DIFFICULTY CLASS: 14                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |
| BUDGET COST: 15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |
| DURATION: 3 turns                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |
| DESCRIPTION<br>Deep packet inspection including protocol reconstruction (rebuilding HTTP streams, email messages, file transfers from packet payloads), malware traffic analysis, and detection of exploitation attempts in traffic. Requires advanced networking and protocol knowledge. What You're Looking For: - Reconstructed HTTP/S traffic (actual data transferred) - Exploitation payloads in network traffic (shellcode, etc) - Malware command protocols (Custom C2 protocols) - Authentication attempts (credentials in transit) - Man-in-the-middle evidence (SSL/TLS downgrade, cert mismatches) - Attacker reconnaissance traffic patterns |        |
| SUCCESS (ROLL ≥ DC 14)<br>Discover TWO Evidence cards from: Exploitation Traffic, C2 Protocol Details, or Attacker Reconnaissance Pattern. Advance (only if no Evidence card produced): Attack Chain +25%, Attribution +25% PARTIAL SUCCESS (roll 12-13); Discover ONE complete evidence + incomplete second. Advance: Attack Chain +15%, Attribution +10% FAILURE (roll < 12): Encryption or obfuscation prevents useful reconstruction.                                                                                                                                                                                                                 |        |
| CHAIN OF CUSTODY<br>PCAP reconstruction must document decoding methodology. Chain of Custody: Moderate (depends on decoding assumptions). / If encrypted traffic decoded, must explain decryption method                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |
| SKILL MODIFIERS<br>+2 if investigator has network forensics certification +2 if protocol reverse engineering experience +1 if NET-01 already completed (building on analysis) -2 if traffic is encrypted and keys not recovered                                                                                                                                                                                                                                                                                                                                                                                                                           |        |
| NOTES<br>• Most detailed network analysis (3 turns) • Requires protocol expertise (HTTP, DNS, custom protocols) • Reveals actual attacker commands and data stolen • Challenging when traffic is encrypted                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |
| INCIDENT ZERO - FORENSICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |

| INVESTIGATION CARD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | MEM-01 |
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| <b>MEMORY DUMP &amp; ANALYSIS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |
| TECHNIQUE: Volatile Memory Forensics (RAM analysis)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |
| MITRE ATTACK: T1015 (Process Injection), T1057 (Process Discovery), T1518 (Software Discovery)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |
| DIFFICULTY CLASS: 13                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |
| BUDGET COST: 15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |
| DURATION: 2 turns                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |
| DESCRIPTION<br>Capture RAM (volatile memory) from running system during incident response, then analyze for active processes, malware, injected code, network connections, and encryption keys in memory. Uses tools like Volatility, Rekall, or proprietary memory analysis suites. What You're Looking For: - Malware processes running in memory - Injected code (shellcode, DLLs in unexpected processes) - Network connections (established C2 connections) - Encryption keys and credentials in memory - Command history from interactive shells - Rootkit or kernel-level hooks |        |
| SUCCESS (ROLL ≥ DC 13)<br>Discover ONE Evidence card from: Active Malware Process, C2 Connection, or Injected Code evidence. Advance (only if no Evidence card produced): Attack Chain +20%, Timeline Completeness +10% PARTIAL SUCCESS (roll 11-12); Discover evidence of suspicious process (incomplete details). Advance: Attack Chain +10%, Timeline Completeness +5% FAILURE (roll < 11): Malware may use anti-forensics in memory; analysis inconclusive                                                                                                                         |        |
| CHAIN OF CUSTODY<br>Memory capture is volatile and must be done immediately. Chain of Custody: Strong if documented with timestamps. / Admissible, but include disclaimer about volatile nature                                                                                                                                                                                                                                                                                                                                                                                        |        |
| SKILL MODIFIERS<br>+2 if investigator trained in memory forensics (Volatility) +1 if malware analysis background +1 if Analyze Evidence action previously discovered malware -2 if memory was overwritten before capture (time-sensitive)                                                                                                                                                                                                                                                                                                                                              |        |
| NOTES<br>• Time-critical: Memory is lost if system is rebooted • Reveals active threats that may not exist on disk • Combines process discovery with malware analysis • Most valuable for finding active C2 connections                                                                                                                                                                                                                                                                                                                                                                |        |
| INCIDENT ZERO - FORENSICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |

| INVESTIGATION CARD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | LOG-02 |
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| <b>DEEP LOG CORRELATION</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |
| TECHNIQUE: Cross-System Log Analysis & Correlation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |
| MITRE ATTACK: T1047 (Account Discovery), T1021 (Remote Services) T1083 (File and Directory Discovery)                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |
| DIFFICULTY CLASS: 13                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |
| BUDGET COST: 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |
| DURATION: 2 turns                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |
| DESCRIPTION<br>Correlate logs from multiple systems (servers, firewalls, IDS, proxies, domain controllers) to build a complete timeline of attacker lateral movement and actions across the environment. Requires SIEM expertise or manual correlation tools. What You're Looking For: - Lateral movement pattern (login on A -> login on B -> etc) - Privilege escalation sequence (user to admin to system) - Command execution across systems - Network connections (firewall - host activity) - Timeline of data access and exfiltration |        |
| SUCCESS (ROLL ≥ DC 13)<br>Discover ONE Evidence card from: Lateral Movement Pattern, Complete Attack Timeline, or Attacker Command Sequence. Advance (only if no Evidence card produced): Timeline Completeness +20%, Attack Chain +25% PARTIAL SUCCESS (roll 11-12); Discover partial timeline (some systems missing logs). Advance: Timeline Completeness +15%, Attack Chain +10% FAILURE (roll < 11): Too many log gaps; timeline cannot be reliably correlated.                                                                          |        |
| CHAIN OF CUSTODY<br>Correlation requires clear documentation of methodology. Chain of Custody: Strong if SIEM tool provided audit trail. / Admissible if correlation process is documented clearly                                                                                                                                                                                                                                                                                                                                           |        |
| SKILL MODIFIERS<br>+2 if investigator has SIEM administration (Splunk, ArcSight) +1 if LOG-01 already completed (building on prior analysis) +1 if detailed explanation of correlation methodology +2 if team provides narrative of suspected attacker movements                                                                                                                                                                                                                                                                             |        |
| NOTES<br>• Most valuable for understanding "how" attacker moved • Reveals attack pace and duration (dwell time) • Can expose failed lateral movement attempts • Requires multiple systems to have logging enabled                                                                                                                                                                                                                                                                                                                            |        |
| INCIDENT ZERO - FORENSICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |

| INVESTIGATION CARD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | MAWM-01 |
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| <b>MALWARE ANALYSIS (DYNAMIC)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |         |
| TECHNIQUE: Behavioral Malware Analysis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |         |
| MITRE ATTACK: T1518 (Software Discovery), T1082 (System Info), T1012 (Query Registry), T1033 (System Owner/User)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |
| DIFFICULTY CLASS: 12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |         |
| BUDGET COST: 15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |         |
| DURATION: 2 turns                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |         |
| DESCRIPTION<br>Execute malware samples in an isolated sandbox environment and record their behavior. Monitor file system changes, registry modifications, network connections, and process creation to understand what the malware does without reverse engineering. Uses tools like Cuckoo, Any.run, or commercial sandboxes. What You're Looking For: - File system changes (what files created/modified) - Registry modifications (persistence mechanisms) - Network communications (DNS, HTTP, etc connections) - Process creation (child processes, injections) - System enumeration (reconnaissance activity) - Anti-analysis techniques (checks for sandbox) |         |
| SUCCESS (ROLL ≥ DC 12)<br>Discover ONE Evidence card from: Malware Behavior Profile, Persistence Mechanism Creation, or C2 Callback Observed. Advance (only if no Evidence card produced): Attack Chain +20%, Attribution +10% PARTIAL SUCCESS (roll 10-11); Malware behavior observed but some details unclear. Advance: Attack Chain +10%, Attribution +5% FAILURE (roll < 10): Malware detects sandbox; exhibits anti-analysis behavior.                                                                                                                                                                                                                         |         |
| CHAIN OF CUSTODY<br>PCAP execution creates video/log recordings of behavior. Chain of Custody: Strong if sandbox logs are preserved. / Admissible (widely accepted malware analysis evidence)                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |         |
| SKILL MODIFIERS<br>+2 if investigator has GREM (GIAC Reverse Engineering Malware) +1 if incident responder with malware analysis training +1 if detailed explanation of behavioral analysis approach -1 if malware implements anti-sandbox techniques                                                                                                                                                                                                                                                                                                                                                                                                               |         |
| NOTES<br>• Safer than static analysis (execution is isolated) • Reveals "what the malware does" not "how it works" • Complements Static Analysis (MAWM-02) well • Useful for identifying persistence and C2 behavior                                                                                                                                                                                                                                                                                                                                                                                                                                                |         |
| INCIDENT ZERO - FORENSICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |         |



| INVESTIGATION CARD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | MALM-02 |
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| <b>MALWARE ANALYSIS (STATIC)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |         |
| TECHNIQUE: Code Reverse Engineering & Analysis<br>MITRE ATTACK: T1140 (Deobfuscate/Deobfuscate Files), T1027 (Obfuscated Files or Information), T1071 (Application Layer Protocol)                                                                                                                                                                                                                                                                                                                                                                                                                                                                |         |
| DIFFICULTY CLASS: 14<br>BUDGET COST: 10<br>DURATION: 2 turns                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |
| DESCRIPTION<br>Disassemble and analyze malware code without execution using reverse engineering tools (IDA Pro, Ghidra, Binary Ninja, etc.). Examine assembly code, strings, imports, and code structure to understand attacker capabilities and techniques. Requires assembly language and debugging expertise. What You're Looking For: - Hardcoded C2 servers, encryption keys - Malware capabilities (spyware, RAT, backdoor, etc.) - Obfuscation techniques (packing, encryption, polymorphism) - Code similarities to known malware families - Exploit codes (zero-days, known CVEs) - Attacker identity clues (developer name, code style) |         |
| SUCCESS (ROLL ≥ DC 14)<br>Discover ONE Evidence card from: Malware Source Code Analysis, Hardcoded C2 Server, or Code Similarity to Known Family. Advance (only if no Evidence card produced): Attack Chain +20% Attribution +25% PARTIAL SUCCESS (roll 12-13): Understand some code features but full analysis incomplete. Advance: Attack Chain +10%, Attribution +10% FAILURE (roll < 12): Malware is heavily obfuscated; code analysis inconclusive.                                                                                                                                                                                          |         |
| CHAIN OF CUSTODY<br>Reverse engineering analysis is documented with screenshots Chain of Custody: Moderate (interpretation-dependent). - Conclusions must be clearly explained for admissibility                                                                                                                                                                                                                                                                                                                                                                                                                                                  |         |
| SKILL MODIFIERS<br>+3 if investigator has GREM (GIAC Reverse Engineering Malware) +2 if assembly language and debugging expertise +1 if MALM-01 already completed (building on behavioral findings) +1 if detailed explanation of reverse engineering approach -2 if malware is polymorphic/heavily obfuscated                                                                                                                                                                                                                                                                                                                                    |         |
| NOTES<br>• Highest skill requirement (DC 14) • Reveals "how the malware works" • Can identify code reuse and attacker patterns • Complements Behavior Analysis (MALM-01) well • Time-consuming (2 turns represents weeks of analysis)                                                                                                                                                                                                                                                                                                                                                                                                             |         |
| INCIDENT ZERO - FORENSICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |         |

| EVIDENCE CARD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | EVD-01 |
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| <b>CREDENTIAL DUMPER MALWARE</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |
| TYPE: Malware & Persistence<br>MITRE ATTACK: T1003 (OS Credential Dumping), T1556 (Modify Auth)                                                                                                                                                                                                                                                                                                                                                                                                                               |        |
| CHAIN OF CUSTODY: STRONG □ - Binary file can be hashed (MD5, SHA-1, SHA-256) - File timestamps document creation/modification - Admissible in court with hash validation                                                                                                                                                                                                                                                                                                                                                      |        |
| DESCRIPTION<br>Malware sample recovered from compromised system that dumps user credentials (SAM file, LSASS process, password hashes, or Kerberos tickets). Examples: Mimikatz, Pdump, LaZagne, where It Was Found: - In System32 directory (hidden with attributes) - In %temp directory (temporary staging) - In admin user AppData (stealth installation) What It Reveals: - Attacker objective: Privilege escalation - Persistence vector: Credential harvesting - Attack phase: Privilege escalation - Lateral movement |        |
| DISCOVERY SOURCE<br>- DISK-01/DISK-02: Found as file artifact - MEM-01/MEM-02: Found as running process in memory - MALM-01: Behavior shows credential dumping actions - MALW-02: Code analysis identifies dumping capabilities                                                                                                                                                                                                                                                                                               |        |
| INVESTIGATIVE LEAD<br>"We found a credential dumper. Let's analyze its behavior (MALW-01) to understand exactly what credentials were captured. Then we can assume those accounts are compromised."                                                                                                                                                                                                                                                                                                                           |        |
| IMPACT ON PROGRESS METERS<br>- Attack Chain: +15% (shows escalation phase) - Attribution: +10% (dumper choice shows attacker sophistication) - Timeline: +10% (timestamp shows when escalation occurred)                                                                                                                                                                                                                                                                                                                      |        |
| FEEDS INTO MODULES<br>- HARDENING: "Implement credential guard to prevent dumping" - NETWORK BUILDING: "Isolate admin credentials in PAM" - AUDIT: "Verify controls around credential access logging"                                                                                                                                                                                                                                                                                                                         |        |
| INCIDENT ZERO - FORENSICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |

| EVIDENCE CARD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | EVD-04 |
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| <b>SUSPICIOUS ADMIN LOGIN (TIMELINE)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |
| TYPE: Credentials & Access<br>MITRE ATTACK: T1078 (Valid Accounts), T1021 (Remote Services), T1550 (Use Alternate Authentication Material)                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |
| CHAIN OF CUSTODY: STRONG □ - Log entry with timestamp and source - digitally signed event log - Corroborated by other log sources - Fully admissible in court                                                                                                                                                                                                                                                                                                                                                                                                                      |        |
| DESCRIPTION<br>Administrator account login with suspicious characteristics: - Unusual time (3 AM instead of business hours) - Unusual location (different country/VPN) - Unusual source (remote desktop instead of VPN) - Batch processing (multiple logins in seconds) Example log entry: 2024-10-18 03:22:15 - User: Administrator Source: 192.0.2.100 (Russia) Protocol: RDP / SSH Success: Yes What It Reveals: - Credential compromise (credentials being used by attacker) - Privilege level compromised (admin account) - Lateral movement likely (attacker on network now) |        |
| DISCOVERY SOURCE<br>- LOG-01: Event Log Analysis shows unusual login event - LOG-02: Correlation across multiple systems - TIMELINE-01: Used to establish attack progression                                                                                                                                                                                                                                                                                                                                                                                                       |        |
| INVESTIGATIVE LEAD<br>"Admin account was compromised. Questions to answer: - When was the password changed? (before or after login?) - What other logins occurred after this? (lateral movement) - Was there any password reset? (attacker covering tracks) - What systems did this account access? (scope of compromise)"                                                                                                                                                                                                                                                         |        |
| IMPACT ON PROGRESS METERS<br>- Attack Chain: +20% (clear escalation point) - Timeline: +25% (login timestamp anchors timeline) - Attribution: +10% (geolocation may hint at attacker origin)                                                                                                                                                                                                                                                                                                                                                                                       |        |
| FEEDS INTO MODULES<br>- HARDENING: "Implement MFA on admin accounts" - NETWORK BUILDING: "Isolate admin access to PAM" - AUDIT: "Review admin account access controls and logging"                                                                                                                                                                                                                                                                                                                                                                                                 |        |
| INCIDENT ZERO - FORENSICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |

| INVESTIGATION CARD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | TIMELINE-01 |
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| <b>TIMELINE RECONSTRUCTION</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |
| TECHNIQUE: Event Correlation & Chronological Analysis<br>MITRE ATTACK: T1074 (Data Staged), T1087 (Account Discovery), T1046 (Network Service Discovery)                                                                                                                                                                                                                                                                                                                                                                 |             |
| DIFFICULTY CLASS: 13<br>BUDGET COST: 5<br>DURATION: 1 turn                                                                                                                                                                                                                                                                                                                                                                                                                                                               |             |
| DESCRIPTION<br>Synthesize evidence from multiple sources (logs, timestamps, file metadata, malware analysis) into a unified chronological timeline of the attack. Identify sequence of events, dwell time, and decision points. What You're Looking For: - Entry point and initial compromise time - Privilege escalation points and timing - Lateral movement sequence - Data reconnaissance timeline - Exfiltration timing (when, how much, for how long) - Dwell time (how long attacker in network before detection) |             |
| SUCCESS (ROLL ≥ DC 13)<br>Discover ONE Evidence card: Complete Attack Timeline with key decision points and transitions between phases identified. Advance (only if no Evidence card produced): Timeline Completeness +25%, Attack Chain +15% PARTIAL SUCCESS (roll 11-12): Partial timeline with some events missing or unclear. Advance: Timeline Completeness +15%, Attack Chain +10% FAILURE (roll < 11): Too many timestamp discrepancies; timeline unreliable.                                                     |             |
| CHAIN OF CUSTODY<br>Timeline must reference source evidence for each event. Chain of Custody: Strong if well-documented and cross-referenced / Admissible if timeline sources are cited                                                                                                                                                                                                                                                                                                                                  |             |
| SKILL MODIFIERS<br>+2 if investigator is DFIR (Digital Forensics & Incident Response) +1 if LOG-01 or LOG-02 already completed +2 if detailed explanation synthesizes multiple evidence sources +1 if team notes discrepancies and explains them                                                                                                                                                                                                                                                                         |             |
| NOTES<br>• Critical for understanding attack progression • Cheap (5 budget) but requires multiple prior investigations • Fast (1 turn) but depends on prior evidence collection • Foundation for narrative reconstruction of incident                                                                                                                                                                                                                                                                                    |             |
| INCIDENT ZERO - FORENSICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |             |

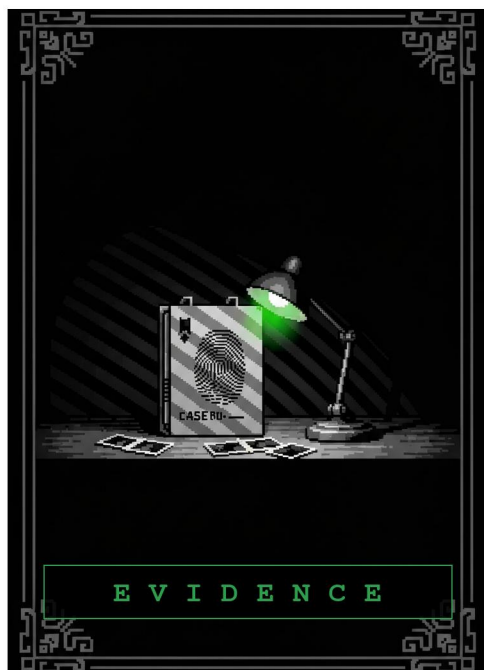
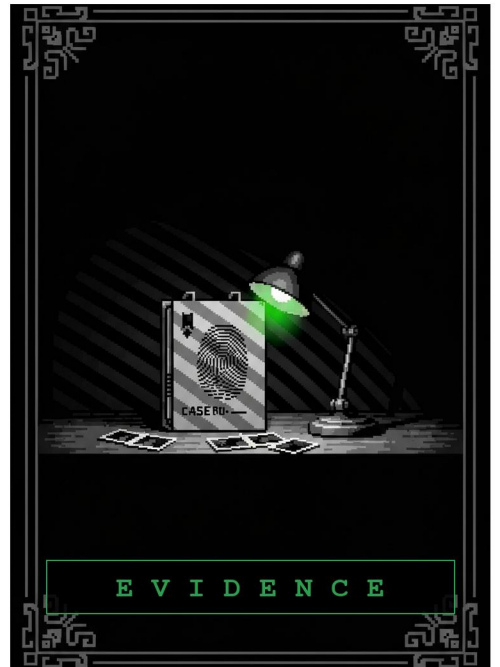
| EVIDENCE CARD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | EVD-02 |
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| <b>COMMAND-AND-CONTROL CALLBACK DOMAIN</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |
| TYPE: Attack Infrastructure<br>MITRE ATTACK: T1071 (Application Layer Protocol), T1573 (Encrypted Channel), T1008 (Fallback Channels)                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |
| CHAIN OF CUSTODY: STRONG □ - Network logs document domain/IP communication - PCAP files timestamp the traffic - DNS logs show query history - Admissible with supporting traffic analysis                                                                                                                                                                                                                                                                                                                                                                                                       |        |
| DESCRIPTION<br>Domain name or IP address that malware communicates with for command and control. Examples: - checkupdate.ru (looks legitimate but is attacker-controlled) - 192.0.2.45 (direct IP address) Where It Was Found: - In malware strings (hardcoded in binary) - In network traffic (outbound connections) - In memory (communication buffers) - In DNS logs (DNS queries) What It Reveals: - Attacker still has access (if domain still active) - C2 Infrastructure operator (may be reused for other campaigns) - Attack sophistication (legitimate-looking domain = higher skill) |        |
| DISCOVERY SOURCE<br>- MALM-01: Dynamic analysis shows C2 connections - MALW-02: Static analysis finds hardcoded domains - NET-01: Network traffic analysis identifies unusual domains - NET-02: Deep packet inspection reconstructs C2 commands                                                                                                                                                                                                                                                                                                                                                 |        |
| INVESTIGATIVE LEAD<br>"We found the C2 domain. Let's do THREAT-01 analysis to: - WHOIS lookup (registrant info) - Historical DNS records (see past resolutions) - Infrastructure mapping (what else is hosted on this IP?) - Passive DNS (VirusTotal, Shodan, etc)"                                                                                                                                                                                                                                                                                                                             |        |
| IMPACT ON PROGRESS METERS<br>- Attack Chain: +15% (confirms persistence vector) - Attribution: +25% (infrastructure links to threat group) - Timeline: +5% (timestamps when C2 was active)                                                                                                                                                                                                                                                                                                                                                                                                      |        |
| FEEDS INTO MODULES<br>- HARDENING: "Block C2 domain via firewall, DNS sinkhole" - NETWORK BUILDING: "Implement egress filtering to C2 ranges" - AUDIT: "Review firewall rules for C2 domain blocking" THREAT INTEL: Can be shared with ISP/CISA for coordinated takedown/blocking.                                                                                                                                                                                                                                                                                                              |        |
| INCIDENT ZERO - FORENSICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |

| EVIDENCE CARD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | EVD-05 |
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| <b>LATERAL MOVEMENT EVIDENCE (PASS-THE-HASH)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |
| TYPE: Lateral Movement<br>MITRE ATTACK: T1550 (Use Alternate Authentication Material), T1110 (Brute Force), T1021 (Remote Services)                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |
| CHAIN OF CUSTODY: MODERATE □ - Hash captured from memory/SAM file - Corroborated by network logs (successful auth events) - Can be cryptographically validated - Admissible with supporting evidence (network logs)                                                                                                                                                                                                                                                                                                                                                           |        |
| DESCRIPTION<br>Evidence that attacker used stolen password hashes to access other systems without knowing the plaintext password. NTLM hash reuse across systems allows lateral movement. What It Shows: - Compromised account: admin-user (hash: A1B2C3456...) - Lateral targets: File server, database server, backup server - Movement pattern: Sequential access across infrastructure What It Reveals: - Attacker sophistication (understanding Windows auth) - Network enumeration (attacker knew what systems exist) - Scope of compromise (multiple systems accessed) |        |
| DISCOVERY SOURCE<br>- LOG-02: Cross-system log correlation shows pattern - NET-01: Network traffic shows auth attempts - MEM-01/MEM-02: Hash visible in memory - DISK-01/DISK-02: SAM file contains hashes                                                                                                                                                                                                                                                                                                                                                                    |        |
| INVESTIGATIVE LEAD<br>"Attacker used pass-the-hash technique. Next steps: - Determine all systems accessed with this hash - Check what actions were taken on each system - Look for privilege escalation or data access on each system"                                                                                                                                                                                                                                                                                                                                       |        |
| IMPACT ON PROGRESS METERS<br>- Attack Chain: +25% (shows sophisticated lateral movement) - Timeline: +15% (timestamps show movement sequence) - Attribution: +15% (technique sophistication shows skill)                                                                                                                                                                                                                                                                                                                                                                      |        |
| FEEDS INTO MODULES<br>- HARDENING: "Implement Credential Guard, Mimikatz mitigations" - NETWORK BUILDING: "Network segmentation to limit lateral move" - AUDIT: "Verify Controls on credential reuse prevention"                                                                                                                                                                                                                                                                                                                                                              |        |
| INCIDENT ZERO - FORENSICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |

| INVESTIGATION CARD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | THREAT-01 |
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| <b>THREAT ATTRIBUTION ANALYSIS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |           |
| TECHNIQUE: Threat Intelligence & Attribution<br>MITRE ATTACK: S8888 group / S8888 software identification Requires synthesis of all prior evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |           |
| DIFFICULTY CLASS: 15<br>BUDGET COST: 20<br>DURATION: 3 turns                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |           |
| DESCRIPTION<br>Synthesize all collected evidence (malware, infrastructure, tactics, timeline, timeline) to attribute the attack to a known threat group, nation-state, or attacker profile. Includes cross-referencing with threat intelligence databases, academic papers, and law enforcement data. This is the highest-level attribution analysis. What You're Looking For: - Similar attacks in CTI databases (VirusTotal, OSINT, etc.) - Malware signatures matching known threat groups - Tactics & Techniques (TTPs) matching known profiles - Infrastructure (domains, IPs) linked to known campaigns - Language/coding style hints about attacker origin - Geolocation clues from timestamps and infrastructure - Victim profile matching known group targeting patterns |           |
| SUCCESS (ROLL ≥ DC 15)<br>Discover ONE Evidence card: Threat Attribution Report with confidence level (60-90%) linking to specific threat group. Advance (only if no Evidence card produced): Attribution Confidence +35%, Attack Chain +10% PARTIAL SUCCESS (roll 13-14): Partial attribution (likely group/profile but not 100% certain) Advance: Attribution Confidence +25%, Attack Chain +5% FAILURE (roll < 13): Insufficient evidence for reliable attribution.                                                                                                                                                                                                                                                                                                            |           |
| CHAIN OF CUSTODY<br>Attribution must cite specific evidence for each finding. Chain of Custody: Moderate (depends on CTI source reliability) - Confidence level must be documented (70% vs. 90% certainty)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |           |
| SKILL MODIFIERS<br>+2 if investigator has threat intelligence background +1 if access to premium CTI services (CrowdStrike, Mandiant) +1 per prior investigation showing strong evidence patterns +2 if detailed narrative synthesizes multiple evidence sources -2 if evidence is sparse or conflicting                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |           |
| CONTINUED IN MODULE RULES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |           |
| INCIDENT ZERO - FORENSICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |           |

| EVIDENCE CARD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | EVD-03 |
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| <b>PERSISTENCE MECHANISM (SCHEDULED TASK)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |
| TYPE: Malware & Persistence<br>MITRE ATTACK: T1043 (Scheduled Task/Job), T1543 (Create/Modify System Process)                                                                                                                                                                                                                                                                                                                                                                                                               |        |
| CHAIN OF CUSTODY: STRONG □ - Task definition stored in %OS (Windows registry/filesystem) - Can be exported and hashed - Creation/modification timestamps available - Fully admissible in court                                                                                                                                                                                                                                                                                                                              |        |
| DESCRIPTION<br>Scheduled task or cron job configured to execute malware at regular intervals (hourly, daily, on system startup). Ensures malware runs even if process is killed or system reboots. Example: - Task: "Windows_Update.Service" (disguised name) - Runs: System startup + every 4 hours - Executes: C:\Windows\System32\cmd.exe (hidden location) What It Reveals: - Attacker skill level (simple but effective) - Intent: Long-term access/persistence - Sophistication: Low-to-medium (persistence is basic) |        |
| DISCOVERY SOURCE<br>- DISK-02: Task files in Windows registry/filesystem - LOG-01: Task execution appears in logs - MEM-01: Task execution visible in running processes - MALM-01: Dynamic analysis shows task creation                                                                                                                                                                                                                                                                                                     |        |
| INVESTIGATIVE LEAD<br>"We found a scheduled task executing malware. Key questions: - When was this task created? (timestamp analysis) - What executable does it run? (acquire and analyze - MALM-01) - Is the executable still present? (filesystem search) - Is the task still active? (persistence threat)"                                                                                                                                                                                                               |        |
| IMPACT ON PROGRESS METERS<br>- Attack Chain: +20% (clearly shows persistence phase) - Timeline: +15% (task timestamps show when persistence installed) - Attribution: +5% (persistence technique is common)                                                                                                                                                                                                                                                                                                                 |        |
| FEEDS INTO MODULES<br>- HARDENING: "Implement AppLocker/code signing for scheduled task executables" - NETWORK BUILDING: "Enable scheduled task logging and analysis" - AUDIT: "Verify controls on scheduled task creation"                                                                                                                                                                                                                                                                                                 |        |
| INCIDENT ZERO - FORENSICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |

| EVIDENCE CARD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | EVD-06 |
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| <b>DATA EXFILTRATION EVIDENCE</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |
| TYPE: Exfiltration<br>MITRE ATTACK: T1020 (Automated Exfiltration), T1030 (Data Transfer Size Limits), T1048 (Exfil Over Alt Protocol)                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |
| CHAIN OF CUSTODY: STRONG □ - Network flow logs (NetFlow, sFlow, or IDS logs) - PCAP files with packet timestamps - Firewall logs documenting outbound connections - Cryptographic hashes of transferred data - Fully admissible in court                                                                                                                                                                                                                                                                                                                                                                    |        |
| DESCRIPTION<br>Evidence of large data transfer from internal network to external attacker-controlled destination. Characteristics: - Volume: 100+ GB transferred in 6-hour window - Timing: During non-business hours (3-6 AM) - Destination: External IP/domain (attacker server) - Protocol: HTTPS, FTP, or custom protocol - Pattern: Consistent data rate (not bandwidth-throttled) What It Reveals: - Scope of compromise (what was accessed) - Attacker objective (data theft vs. ransomware) - Attack timeline (when exfiltration occurred) - Attacker infrastructure (location of receiving server) |        |
| DISCOVERY SOURCE<br>- NET-01: Network traffic analysis shows volume anomalies - NET-02: Packet inspection shows data being transferred - LOG-02: Firewall/proxy logs show external connections - MALM-01: Dynamic analysis shows file staging before exfil                                                                                                                                                                                                                                                                                                                                                  |        |
| INVESTIGATIVE LEAD<br>"Massive data exfiltration detected. Critical questions: - Exactly which files/databases were exfiltrated? - How many customer records are affected? - Can we identify specific data types stolen? - Is the data still being transferred (ongoing threat)?"                                                                                                                                                                                                                                                                                                                           |        |
| IMPACT ON PROGRESS METERS<br>- Attack Chain: +20% (confirms attacker objectives) - Timeline: +20% (exfil duration/timing) - Attribution: +10% (exfil infrastructure may be reused)                                                                                                                                                                                                                                                                                                                                                                                                                          |        |
| FEEDS INTO MODULES<br>- HARDENING: "Data loss prevention (DLP) controls" - NETWORK BUILDING: "Egress filtering, traffic inspection" - DISASTER RECOVERY: "Breach notification scope (data volume)" - AUDIT: "Data protection controls and encryption review"                                                                                                                                                                                                                                                                                                                                                |        |
| INCIDENT ZERO - FORENSICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |



| EVIDENCE CARD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | EVD-07 |
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| <b>ATTACKER INFRASTRUCTURE MAP</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |
| TYPE: Attack Infrastructure                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |
| MITRE ATTACK: Related to C2 infrastructure and command channels                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |
| CHAIN OF CUSTODY: MODERATE <input type="checkbox"/> - WHOIS records are public but can be modified - Historical DNS data from passive DNS services - Correlations need cross-referencing - Admissible with supporting evidence (traffic logs)                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |
| DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |
| Connected map of attacker-controlled infrastructure including multiple domains, IP addresses, registrars, and services. Example Infrastructure Web: - Primary C2: checkupdate.ru (IP: 192.8.2.45) - Alternate C2: update-service.xyz (IP: 192.0.2.46) - Malware hosting: files.example.net (IP: 192.0.2.47) - Registrant: All registered via registrar.ru - ASN: AS64512 (Ukrainian ISP network) What It Reveals: - Attacker operational security (multiple infrastructure) ■ - Attacker resources (ISP relationships, hosting account) - Attacker location hints (registrar, ASN, geo-location) - Attack history (domains registered months/years earlier) - Other campaigns (infrastructure reused for other attacks) |        |
| DISCOVERY SOURCE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |
| - THREAT-01: Threat attribution analysis connects domains - MALW-02: Static analysis finds hardcoded backup domains - NET-01: Network traffic shows multiple C2 attempts - CTI research: VirusTotal, Shodan, Passive DNS services                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |
| INVESTIGATIVE LEAD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |
| "We've mapped attacker infrastructure. Next steps: - Search threat intelligence databases for this infrastructure - Look for connections to known threat groups - Check if infrastructure used in other campaigns - Contact registrar and hosting for takedown - Report to ISP for blocking/monitoring"                                                                                                                                                                                                                                                                                                                                                                                                                 |        |
| IMPACT ON PROGRESS METERS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |
| - Attribution: +30% (infrastructure often linked to groups) - Attack Chain: +15% (understanding attacker preparation) - Timeline: +10% (infrastructure registration dates)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |
| FEEDS INTO MODULES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |
| - HARDENING: "Block all known C2 infrastructure via firewall" - AUDIT: "Threat intelligence integration for blocking" - THREAT INTEL: Shareable with industry, law enforcement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |
| INCIDENT ZERO · FORENSICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |

| EVIDENCE CARD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | EVD-10 |
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| <b>MALWARE BEHAVIOR PROFILE</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |
| TYPE: Malware & Persistence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |
| MITRE ATTACK: Multiple TTPs based on observed behavior                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |
| CHAIN OF CUSTODY: STRONG <input type="checkbox"/> - Sandbox execution video/logs document behavior - Timestamps and sequence recorded - Reproducible analysis methodology - Widely accepted malware analysis evidence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |
| DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |
| Complete profile of malware capabilities and behavior based on dynamic analysis in sandbox environment. Profile Contents: - File system interactions (creates, modifies, deletes) - Registry modifications (persistence mechanisms) - Process creation (parent-child relationships) - Network communications (DNS queries, HTTP requests, IPs) - API calls (Windows/Linux API usage) - Anti-analysis techniques (sandbox evasion) Example Output: - Name: comhost.exe (masquerading as Windows process) - Creates files: C:\Users\A\AppData\Local\Temp\app.exe - Registry: HKLM\Software\Microsoft\Windows\CurrentVersion\Run (persistence) - Network: Connects to update.badsite.ru:443 every 15 minutes - Capabilities: Credential harvesting, File encryption, C2 What It Reveals: - Complete malware capabilities - Attacker operational techniques - Threat level (spyware vs. ransomware vs. trojan) - Indicators of Compromise (IOCs) |        |
| DISCOVERY SOURCE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |
| - MALW-01: Dynamic sandbox analysis produces full profile - MALW-02: Static analysis validates observed behaviors - Combined: Behavior validated against code confirms accuracy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |
| INVESTIGATIVE LEAD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |
| "We have the complete malware profile. Now we can: - Search for all instances of this malware - Hunt for C2 communications (IPs, domains) - Search for created files and artifacts - Link to other malware families (code similarities)"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |
| IMPACT ON PROGRESS METERS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |
| - Attack Chain: +20% (understand capabilities + understand threat) - Attribution: +15% (malware signatures match known families) - Timeline: +10% (behavior timing shows operation phase) - Chain of Custody: +10% (sandbox logs are strong evidence)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |
| FEEDS INTO MODULES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |
| - HARDENING: "Controls to prevent malware execution" - NETWORK BUILDING: "Detection of malware C2 behaviors" - AUDIT: "EDR/SIEM coverage for malware detection"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |
| INCIDENT ZERO · FORENSICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |

| EVIDENCE CARD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | FIND-01 |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |         |
| <b>THREAT INTELLIGENCE REPORT</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |         |
| TYPE: Findings/Conclusions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |         |
| TRIGGERED WHEN: Attribution Confidence ≥ 70%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |
| FINDING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |         |
| Attack attributed to [Threat Group Name] Confidence Level: [60-90% based on evidence] Associated Techniques: [MITRE ATT&CK TTPs] Previous Targets: [Industries/organizations previously targeted] Likely Motivation: [Financial gain, espionage, etc] RECOMMENDATIONS: 1. Notify law enforcement (FBI, Interpol if international) 2. Share intelligence with industry ISACs 3. Monitor for indicators of re-engagement 4. Implement defenses targeting group's known TTPs FEEDS INTO MODULES: → HARDENING: "Defense-in-depth against attributed group" → AUDIT & COMPLIANCE: "Threat model update with attributed group" → INCIDENT RESPONSE: "Playbook for future incidents from group" |         |
| INCIDENT ZERO · FORENSICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |         |

| EVIDENCE CARD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | EVD-08 |
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| <b>ENCRYPTION KEYS RECOVERED</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |
| TYPE: Malware & Persistence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |
| MITRE ATTACK: T1140 (Encrypt/Decrypt), T1552 (Unsecured Credentials), T1074 (Data Staged)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |
| CHAIN OF CUSTODY: MODERATE <input type="checkbox"/> - Keys extracted from memory/binary must be documented - Extraction methodology must be explained - Cross-referencing with code/behavior confirms validity - Admissible with supporting analysis documentation                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |
| DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |
| Encryption keys recovered from memory, disk, or malware code that allow decryption of: - Malware traffic (C2 communications) - Stolen data archives (what was exfiltrated) - Attacker staging servers (accessing their infrastructure) - Backdoor communications (understanding commands) Examples: - AES-256 key found in malware binary - RSA key in process memory (used for C2) - TLS certificates for backdoor listener - Steganography keys (hidden messages in files) What It Reveals: - Encryption strength (military-grade vs. basic obfuscation) - Attacker sophistication (poor key management = careless) - What data can be decrypted (scope of analysis) - Backdoor capabilities (understanding command set) |        |
| DISCOVERY SOURCE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |
| - MEM-02: Deep memory analysis finds encryption keys - DISK-02: File carving recovers keys from slack space - MALW-02: Static analysis finds hardcoded keys - MALM-01: Dynamic analysis reveals keys generated at runtime                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |
| INVESTIGATIVE LEAD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |
| "We recovered encryption keys! This is huge because: - We can decrypt C2 communications (see commands sent) - We can decrypt malware archives (understand what was stolen) - We can access attacker staging servers (more evidence) - We can build stronger attribution (command content)"                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |
| IMPACT ON PROGRESS METERS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |
| - Attack Chain: +25% (understand full communication) - Attribution: +20% (commands reveal attacker objectives) - Timeline: +15% (command history shows action sequence) - Chain of Custody: +15% (encryption is strong evidence)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |
| FEEDS INTO MODULES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |
| - HARDENING: "Secure key management practices" - AUDIT: "Encryption and key management controls" - THREAT INTEL: Keys shared with law enforcement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |
| INCIDENT ZERO · FORENSICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |

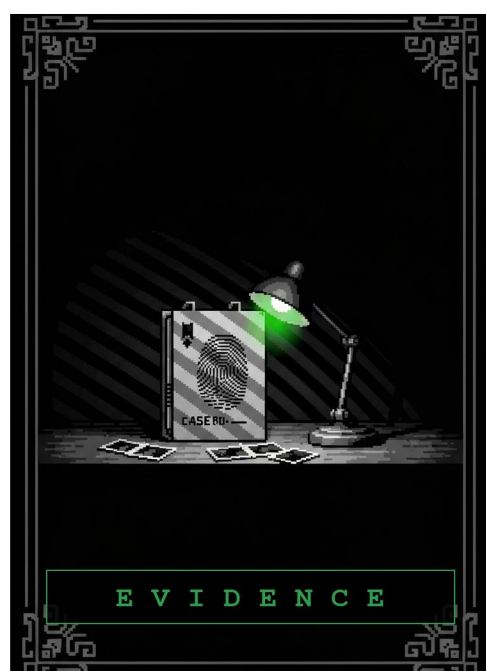
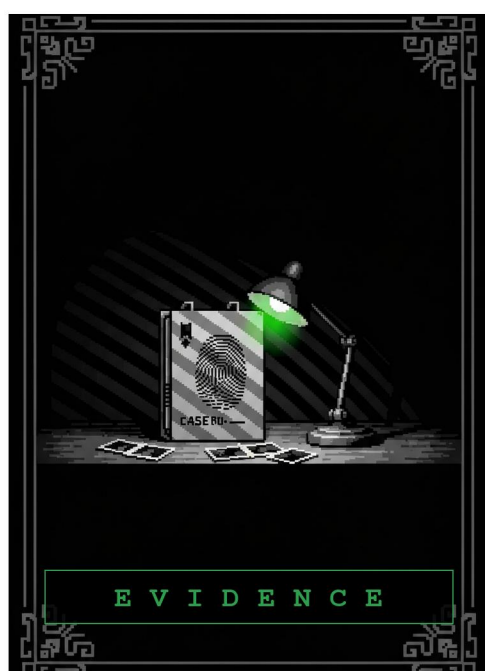
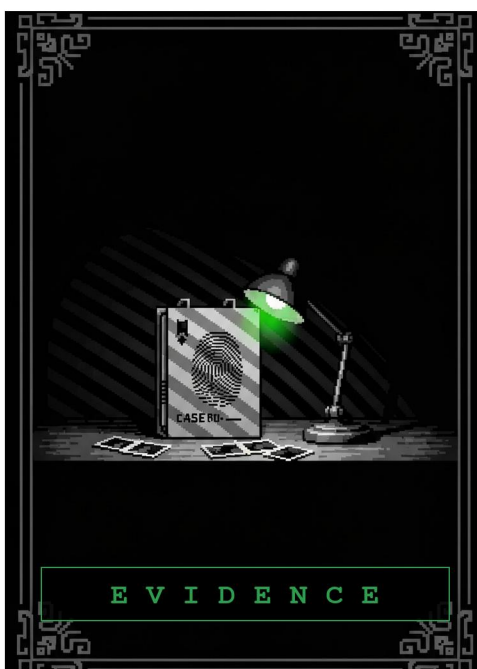
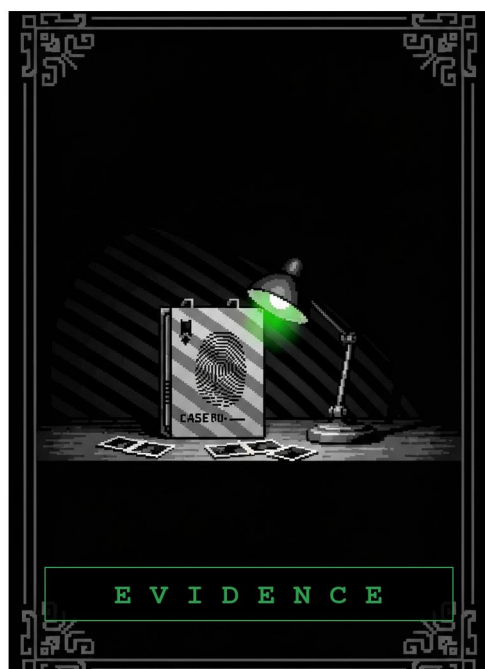
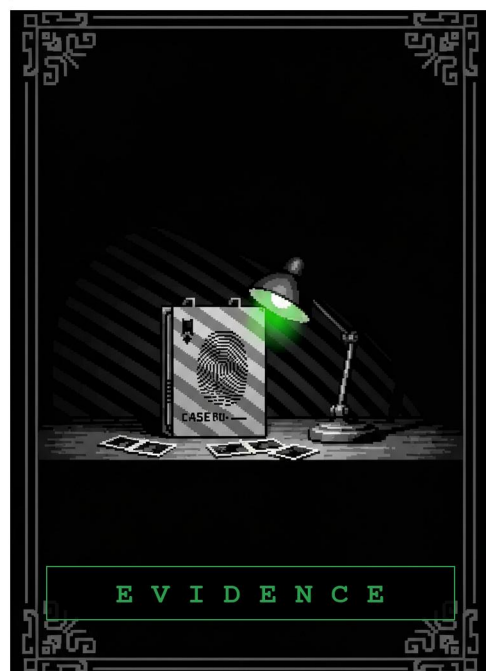
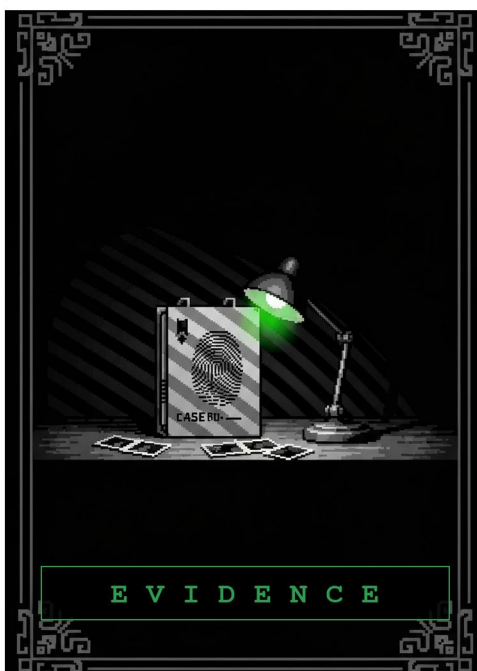
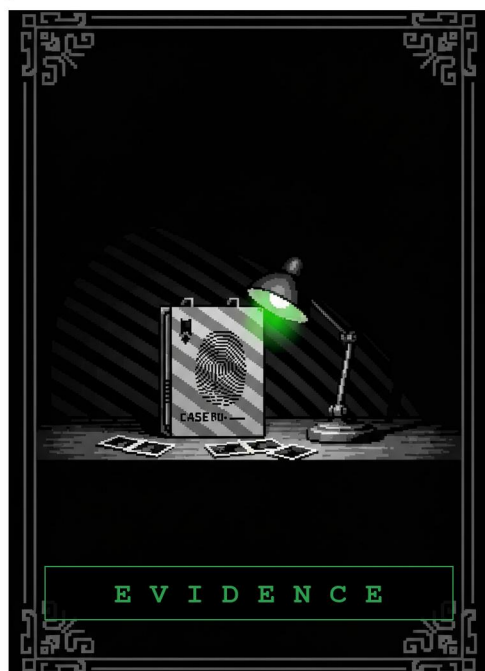
| EVIDENCE CARD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | EVD-11 |
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| <b>FILE STAGING ARTIFACTS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |
| TYPE: Attack Activity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |
| MITRE ATTACK: T1074 (Data Staged), T1005 (Data from Local System)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |
| CHAIN OF CUSTODY: STRONG <input type="checkbox"/> - File hashes document the staging - File timestamps show staging timeline - File content confirms what was staged - Fully admissible in court                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |
| DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |
| Evidence of attacker staging files before exfiltration. Files are collected in a temporary location, compressed, encrypted, then transferred to attacker server. Artifacts Found: - Compressed archives (RAR, 7z, ZIP files) - Partially deleted files (overwrite artifacts) - File lists (text files naming what to steal) - Batch scripts (automated collection scripts) - Temporary directories with suspicious contents Example: - C:\Staging\data_backup.7z (500 MB) - C:\Staging\files_to_get.txt (list of target files) - C:\Staging\collect.bat (automated collection script) What It Reveals: - Data that was targeted (from .txt lists) - Volume of exfiltration (archive size) - Compression ratio (how much data actually stolen) - Attacker knowledge (knew where sensitive data was) |        |
| DISCOVERY SOURCE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |
| - DISK-01/DISK-02: Staging artifacts on disk - LOG-02: Batch script execution in logs - MALM-01: Dynamic analysis shows staging process - NET-01: File transfer evidence (connection to staging dir)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |
| INVESTIGATIVE LEAD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |
| "Attacker staged specific files. This shows: - Exact data that was targeted (from staging lists) - Attack planning (targeted vs. random) - Data sensitivity (what did they prioritize) - Precision of attack (narrow vs. broad data grab)"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |        |
| IMPACT ON PROGRESS METERS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |
| - Attack Chain: +15% (staging phase evidence) - Timeline: +20% (staging timestamps show prep phase) - Attribution: +10% (precision shows targeting sophistication) - Chain of Custody: +10% (file evidence is strong)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |
| FEEDS INTO MODULES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |
| - HARDENING: "Data identification and protection (DLP)" - AUDIT: "Data classification and access controls" - NOTIFICATION: "Specific data breach notification"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |        |
| INCIDENT ZERO · FORENSICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |

| EVIDENCE CARD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | FIND-02 |
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| <b>ATTACK SURFACE ANALYSIS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |         |
| TYPE: Findings/Conclusions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |         |
| TRIGGERED WHEN: Attack Chain ≥ 75%                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |
| FINDING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |         |
| Entry Point: [Method used for initial compromise] Exploited Vulnerability: [CVE, weak auth, configuration gap] Escalation Point: [Where privilege escalation occurred] Lateral Movement Paths: [Systems accessed after pivot] ROOT CAUSE: - [Patch missing, configuration weakness, process gap] RECOMMENDATIONS: 1. Patch entry-point vulnerability immediately 2. Implement detection for exploitation attempts 3. Restrict lateral movement (network segmentation) 4. Update architecture to prevent this attack path FEEDS INTO MODULES: → HARDENING: "Specific technical hardening measures" → NETWORK BUILDING: "Architecture redesign to block attack path" → AUDIT: "Control gap remediation" |         |
| INCIDENT ZERO · FORENSICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |

| EVIDENCE CARD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | EVD-09 |
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| <b>ATTACKER COMMAND HISTORY</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |
| TYPE: Attack Activity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |
| MITRE ATTACK: T1059 (Command & Scripting Interpreter), T1059.001 (PowerShell)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |
| CHAIN OF CUSTODY: STRONG <input type="checkbox"/> - Command history from shell/terminal logs - PowerShell transcript logs (if enabled) - Memory forensics shows running command buffer - Timestamps document command execution order - Fully admissible in court                                                                                                                                                                                                                                                                                                                                                                                             |        |
| DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |
| Recovered history of commands executed by attacker on compromised systems. Shows attacker's actions, objectives, and decision-making process. Examples: - PowerShell: Get-AdUser -Filter *   Export-CSV C:\temp\ad.csv - CMD: dir \\backup-server\share - Bash: find / -name "*.sql" -> -name "*.db" 2>/dev/null What It Reveals: - Attacker objectives (looking for what? ad users? databases?) - Attacker knowledge (familiar with Windows/Linux/networks) - Attack sophistication (script-kiddie vs. skilled operator) - Targeting specificity (random exploration vs. targeted search) - Timeline of activities (sequence of commands shows progression) |        |
| DISCOVERY SOURCE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |
| - MEM-02: Memory forensics finds recent command buffer - LOG-02: Command execution logging (PowerShell, bash history) - DISK-01: Shell history files (.bash_history, PowerShell logs) - MALW-01: Dynamic analysis shows commands sent to shell                                                                                                                                                                                                                                                                                                                                                                                                               |        |
| INVESTIGATIVE LEAD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |
| "We have the attacker's command history! This shows us: - What systems they were looking for - What data they searched for - How much time they spent on each system - When they pivoted to new systems - When they started exfiltration - If they set up backdoors or persistence"                                                                                                                                                                                                                                                                                                                                                                          |        |
| IMPACT ON PROGRESS METERS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |
| - Timeline: +20% (command timing shows exact sequence) - Attack Chain: +25% (command progression shows phases) - Attribution: +15% (command style/language hints) - Chain of Custody: +10% (strong admissible evidence)                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |
| FEEDS INTO MODULES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |
| - HARDENING: "PowerShell transcript logging, command audit" - AUDIT: "Verify logging of command execution" - TRAINING: "Identify what commands should have triggered alerts"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |
| INCIDENT ZERO · FORENSICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |

| EVIDENCE CARD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | EVD-12 |
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| <b>ANTI-FORENSICS EVIDENCE</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |
| TYPE: Attack Activity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |
| MITRE ATTACK: T1070 (Indicator Removal), T1485 (Data Destruction), T1556 (Modify Authentication Process)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |
| CHAIN OF CUSTODY: MODERATE <input type="checkbox"/> - Evidence is lack of evidence (absences are hard to prove) - Comparison with known baselines shows anomalies - Log deletion tools detected and documented - Admissible with supporting context (other evidence)                                                                                                                                                                                                                                                                                                                                                                                                                                                           |        |
| DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |
| Evidence that attacker actively tried to cover their tracks using anti-forensics techniques. Anti-forensics Found: - Event log deletion (Clear-EventLog PowerShell) - File timestamp manipulation (TimeStomp) - Log overwriting (dd commands filling logs) - File shredding (secure deletion of evidence) - Registry clearing (Cleanmgr, Cleaner, etc) - Malware self-deletion after execution What It Reveals: - Sophistication (advanced attackers use anti-forensics) - Awareness (attacker knew forensics would be used) - Intent (intentional cover-up vs. accidental trail) - What they're hiding (deleted logs - they knew activities would be suspicious) - Attack planning (anti-forensics in playbook = pre-planned) |        |
| DISCOVERY SOURCE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |
| - LOG-01/LOG-02: Gaps in logs (suspicious absences) - DISK-01/DISK-02: Deleted log files - Anti-forensic tools - MEM-01/MEM-02: Anti-forensic process running in memory - MALW-01: Dynamic analysis shows self-deletion - MALW-02: Code analysis finds anti-forensic capabilities                                                                                                                                                                                                                                                                                                                                                                                                                                              |        |
| INVESTIGATIVE LEAD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |
| "Attacker used anti-forensics. This actually helps because: - Proves attacker sophistication (means skilled opponent) - Indicates intentional harm (not accidental) - Suggests what they're hiding (what logs were deleted) - Helps attribution (anti-forensics technique is signature) - Can reconstruct from other sources (memory, network logs)"                                                                                                                                                                                                                                                                                                                                                                           |        |
| IMPACT ON PROGRESS METERS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |
| - Attribution: +20% (anti-forensic technique is signature) - Attack Chain: +10% (shows post-attack phase) - Timeline: -10% (anti-forensics makes timeline harder) - Chain of Custody: +5% (proves intentional cover-up)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |        |
| FEEDS INTO MODULES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |
| - HARDENING: "Immutable logging (cloud, WORM storage)" - NETWORK BUILDING: "Centralized log aggregation" - AUDIT: "Log integrity and anti-tampering controls"                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |
| INCIDENT ZERO · FORENSICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |

| EVIDENCE CARD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | FIND-03 |
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| <b>PERSISTENCE MECHANISMS DISCOVERED</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |         |
| TYPE: Findings/Conclusions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |
| TRIGGERED WHEN: Multiple persistence artifacts found                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |         |
| FINDING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |
| Primary Persistence: [Scheduled task, registry run, etc] Backup Persistence: [Redundant persistence methods] Dormancy: [How long could malware remain active undetected] THREAT: Attacker likely still has access (persistence remains active) - Malware calls home regularly (C2 connections) - Can re-establish access if initial access closed - May deploy additional payloads over time IMMEDIATE ACTIONS: 1. Fully remediate all discovered persistence mechanisms 2. Search for backup persistence (often multiple methods) 3. Monitor for re-establishment of access 4. Assume attacker may have staged additional backdoors FEEDS INTO MODULES: → HARDENING: "Persistence prevention and detection" → DISASTER RECOVERY: "Scope of remediation (how deep?)" → AUDIT: "Endpoint protection review" |         |
| INCIDENT ZERO · FORENSICS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |         |



INVESTIGATIVE GAPS & RECOMMENDATIONS

TYPE: Findings/Conclusions

TRIGGERED WHEN: Investigation completes (Victory or Failure)

FINDING

Key Questions Answered: - [ ] Attack entry point identified? - [ ] Attacker motivation understood? - [ ] Threat actor identified (attribution)? - [ ] Data compromised: volume and sensitivity? - [ ] Current access status: eliminated or ongoing? - [ ] Persistence mechanisms: removed or active? Remaining Questions: - [List specific unknowns from the investigation] - [What evidence gaps prevent complete understanding] - [What would close these gaps (more investigation, experts)] NEXT STEPS: 1. [If gaps remain: External forensics firm for deep analysis] 2. [Law enforcement involvement for attribution/prosecution] 3. [Threat intelligence: share findings with industry] 4. [Lessons learned: update hardening/network architecture] FEEDS INTO MODULES: -> AUDIT & COMPLIANCE: "Post-incident review and control updates" -> TRAINING: "Lessons learned session with all teams" -> STRATEGIC: "Investment in detection/response capabilities"

